National University of Computer and Emerging Sciences
Islamabad Campus

Digital Logic Design (EE1005)

Sessional-I Exam

Total Time (Hrs): 1
Total Marks: 40
Total Questions: 5

Date: Feb 23, 2026

Course Instructor(s):
Mr. Junaid Ali Khan, Ms. Naveen Khan, Ms. Laraib Afzal, Ms. Samia Aziz, Ms. Maryam Sana

Section(s): SE (A, B, C), CY (A,B,C), DS (A,B), AI (A,B,C)

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Roll No **Course Section** **Student Signature**

Do not write below this line.

Attempt all the questions.

1. ***Calculator is allowed during the exam.***
2. ***All steps must be shown clearly. Direct answers or missing steps will receive zero marks even if the final answer is correct.***
3. ***Write all final answers clearly in the provided answer space. Only clean and clearly written solutions will be considered for marking.***
4. ***Use the extra sheet for the rough. Answers written in the rough area will not be marked.***
5. ***Attempt all questions neatly and label expressions, circuits and steps properly to receive full marks.***
6. ***Any form of cheating or unauthorized assistance will result in cancellation of the paper.***
7. ***Over writing, cutting and use of pencil will lead to zero marks***
8. ***Rules/laws of boolean algebra are given at end page***
9. ***Attempt Q4 and Q5 on answer sheet***

Attempt all the questions.

[CLO 1: Understand the foundations of digital logic, number systems and basic circuit concepts]

Q1: Convert the 8-bit **gray code** to its equivalent **BCD (Binary-Coded Decimal)** representation.
[5 marks]

11010110

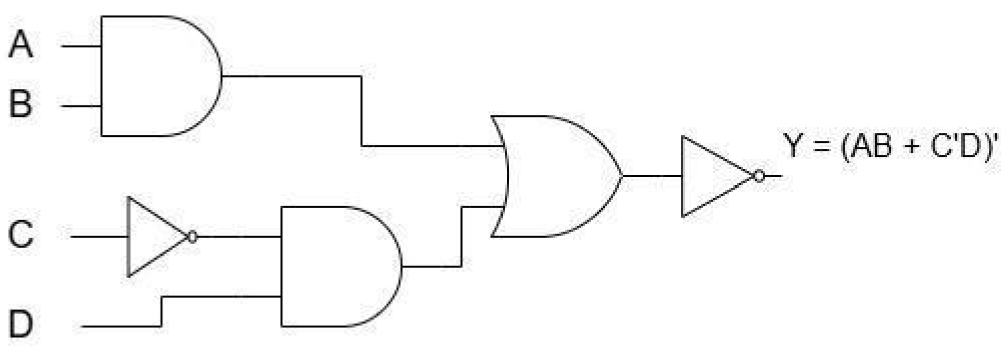
Sol

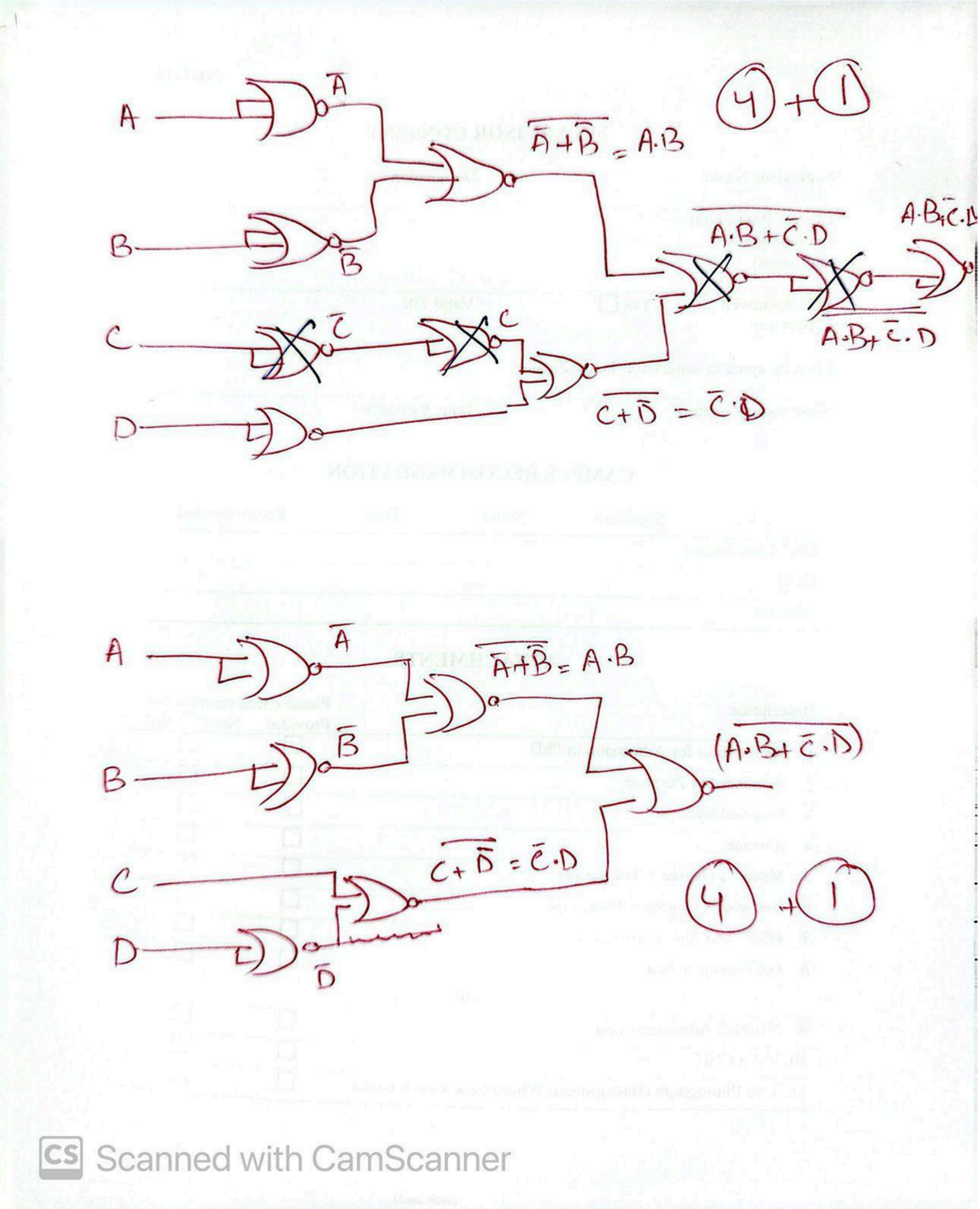
11010110
Gray → Binary:
Binary: 10011011
Binary → Decimal:
10011011 = 128+16+8+2+1
Decimal: 155
Decimal → BCD
BCD: 0001 0101 0101

Q2: Perform the addition of given numbers using BCD. **[5 Marks]**

1	1	1	
2	4	8	6
+ 3	6	9	7
0110	**1011**	**10010**	**1101**
	0110	**0110**	**0110**
0110	**0001**	**1000**	**0011**

Q3: Convert the given AOI circuit into an equivalent circuit using only NOR gates. Simplify the circuit and redraw the final NOR-only implementation.
(Show all conversion steps clearly) [4+4+2=10 marks]





Attempt Q4 and Q5 on answer sheet

[CLO 2: Demonstrate the acquired knowledge to apply techniques related to the design and

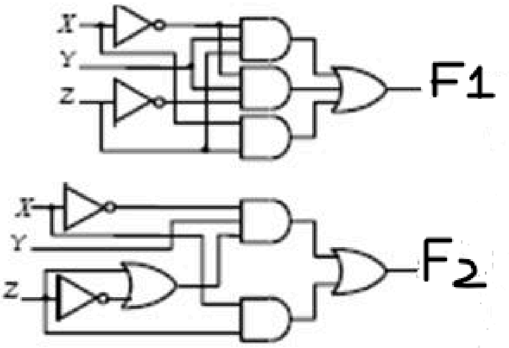
analysis of digital electronic circuits including boolean algebra and multi variable karnaugh map methods(2)(3)]

Q4: Convert the given Boolean function into its standard form (Canonical Sum of Product). Derive the final equation, make a truth table and write the minterms like (m0,m1….) in it. **[4+3=7marks]**

F (A, B, C) = AB+BC+AC

Q5: Perform the analysis of the given circuit below **[4+2+3+4=13 Marks]**

a. Find Boolean expressions for **F1 and F2**.
b. You must Simplify both(F1 and F2) Boolean expressions using Boolean algebra rules.
c. Make sure to verify via the truth table**(F1=F2=simplified expression)**.
d. Redraw the circuit for **simplified expression** using ***NAND*** gate only.



Algebraic Rules/laws

Postulate 2	(a)	$x + 0 = x$	(b)	$x \cdot 1 = x$
Postulate 5	(a)	$x + x' = 1$	(b)	$x \cdot x' = 0$
Theorem 1	(a)	$x + x = x$	(b)	$x \cdot x = x$
Theorem 2	(a)	$x + 1 = 1$	(b)	$x \cdot 0 = 0$
Theorem 3, involution		$(x')' = x$		
Postulate 3, commutative	(a)	$x + y = y + x$	(b)	$xy = yx$
Theorem 4, associative	(a)	$x + (y + z) = (x + y) + z$	(b)	$x(yz) = (xy)z$
Postulate 4, distributive	(a)	$x(y + z) = xy + xz$	(b)	$x + yz = (x + y)(x + z)$
Theorem 5, DeMorgan	(a)	$(x + y)' = x'y'$	(b)	$(xy)' = x' + y'$
Theorem 6, absorption	(a)	$x + xy = x$	(b)	$x(x + y) = x$